CohortCert: Shared Evidence and Proof Costs for Cohort Decisions

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***Abstract*—Comparative text analytics compares rates of semantic properties, such as toxicity, across groups of records. A request can ask several comparisons, equivalence judgments or rankings over overlapping groups. Obtaining a label may require an expensive model call, yet sharing labels alone does not identify which reads will justify every requested answer within budget. We view execution as completing a decision certificate over shared evidence, and distinguish missed acquisition opportunities from proof-cost barriers under fixed inference rules. CohortCert compiles comparisons into signed shared counts, reuses anytime-valid count intervals, and expresses complete-request certification as a compact optimization program. Online acquisition uses observed evidence; independently checked offline bounds quantify proof costs on realized sampling paths. Across $73{,}760$ planned saved-label replays, DifficultyGreedy improves CivilComments completion over WidthGreedy by 0.78 and 1.16 percentage points for crowd and model labels, while GoEmotions nearly ties. Forecast is less effective than the simpler heuristic. The bounds construct within-budget proofs in 157 of 192 preselected cases and establish a signed-overlap proof-cost benefit in 30; these are conditional hindsight findings. Seven configured vector-label programs and 23 live query executions further examine inference choices and actual model costs.**

***Index Terms*—semantic query processing, finite-population inference, Boolean certificates, evidence acquisition, cohort comparison**

I. Introduction

Suppose an analyst wants to compare complaint rates in two overlapping groups of comments. The request asks three questions: is group A's rate more than five percentage points above B's, is it less than six points above, and are the two rates within ten points? A model can label individual comments, but a fixed budget prevents labeling everything. The system must choose which comments to inspect and justify all three answers.

Figure 1 gives a small example. The observations support a rate difference between 0.07 and 0.21. The first question is therefore true and the second false; the third is still unresolved. A false answer is a legitimate result for one question. It does not eliminate the need to answer the others. We call the request *complete* only when every requested decision is justified. A strict top-k request must also identify the selected groups.

Sharing matters at two levels. First, a comment in both groups is one physical record, so we should acquire its label once and use its exact contribution to both rates. Second, the resulting evidence can settle several requested decisions. Treating each group or question in isolation loses this structure.

An unfinished request raises another useful question: did the allocator overlook a completing set of reads, or would the current inference rules require more than the budget? Observed execution alone cannot distinguish these explanations. CohortCert adds an offline cost analysis that can. Given the realized sampling paths, it constructs a completing allocation or proves a lower bound on its cost. If its bounds are too wide, the case remains inconclusive. This analysis identifies where allocation has room to improve and where a different inference rule may be needed.

Existing semantic engines optimize expensive model operators [1]–[3], while Evergreen verifies aggregate and ordinal claims [4]. Boolean evaluation [5]–[7] and sequential inference [8]–[11] supply relevant foundations. Our object is the *complete comparative request*: its shared evidence, the decisions that evidence supports, and the acquisition cost of completing them under stated rules.

CohortCert has three parts. A *compiler* expresses the questions using shared counts. An *executor* selects records, updates count intervals, and checks decisions. An *offline evaluator* uses full realized paths to bound the least evidence needed. Online allocators never receive those future labels; Forecast uses hypothetical paths instead. This distinction is essential when interpreting the results.

We contribute a complete-request contract over shared signed counts, a compact completion program with checked cost bounds, and matched evaluations that isolate overlap propagation and allocation across decisions. The separate Technical Companion retains full derivations and reproduction parameters. Code, data manifests, the current paper and the

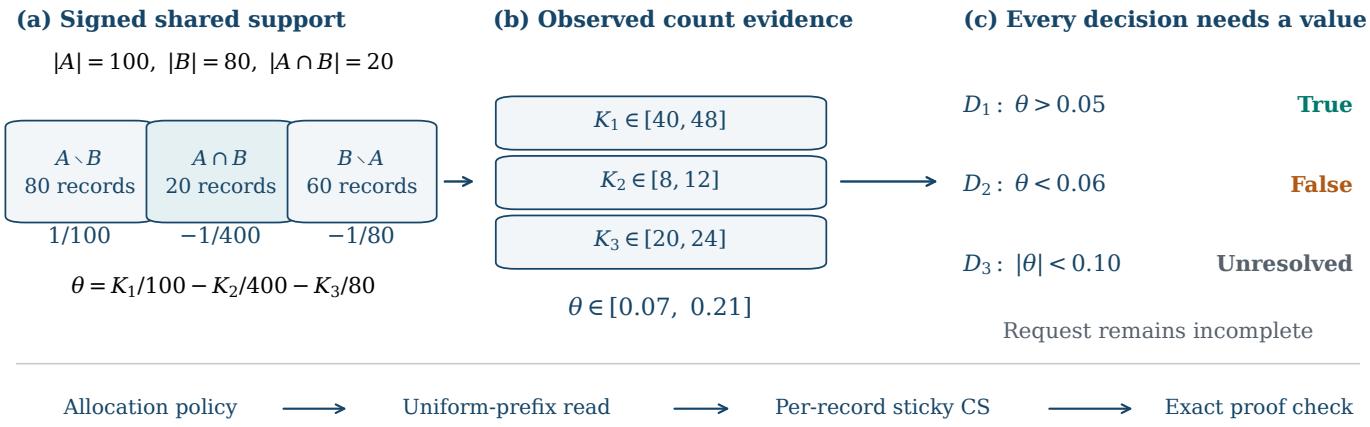


Fig. 1. Running example, using illustrative values. The three groups of records are $A \setminus B$, $A \cap B$, and $B \setminus A$; K_1, K_2, K_3 count their positive labels. Shared count intervals imply $0.07 \le \theta \le 0.21$ for the rate difference. This proves D_1 true and D_2 false, but does not settle D_3. The loop at the bottom reads more records and updates the same evidence until all decisions are settled or the budget is spent.

companion are publicly available.[1]

II. From Shared Records to Justified Answers

A. What the Request Means

A request fixes texts, known cohorts, one binary label operator, thresholds, a call budget B and risk α. A positive label is $y_i = 1$. For cohort G_g of known size n_g, its rate is the fraction of positive records, $\mu_g = \sum_{i \in G_g} y_i / n_g$. The operator is versioned: switching from crowd annotations to model labels changes the target rate.

Each comparison/equivalence returns true, false or unresolved; a completed ranking returns its set. Evidence and cost accompany outputs. A primary-study call is one submitted record–predicate label attempt, including pilot and failed attempts. Within a request, the same label serves every relevant comparison. Other requests and predicates are charged afresh. Cohort metadata are given; text-feature preparation and physical model time are accounted separately.

B. Why One Shared Count Is Enough

In Fig. 1, A has 100 records, B has 80, and their intersection has 20. A positive record in the intersection increases both rates, but increases B's more because B is smaller. Its net contribution is $1/100 - 1/80 = -1/400$. Thus the difference depends on only three positive counts: K_1 in A only, K_2 in the intersection, and K_3 in B only:

$$\theta = \mu_A - \mu_B = K_1/100 - K_2/400 - K_3/80. \qquad (1)$$

From the displayed count ranges, the smallest possible difference is $40/100 - 12/400 - 24/80 = 0.07$; the largest is $48/100 - 8/400 - 20/80 = 0.21$. For a negative coefficient we use the *upper* count to obtain the lower difference, and conversely. This is all that *signed propagation* means.

[1]https://github.com/yzd2002/CohortCert. The original evidence archives are fixed at commit `aae1e7b`; the repository manifests identify the current manuscript and Technical Companion.

TABLE I
The objects behind the formulas. A count interval is evidence; a decision certificate uses that evidence; a cost bound concerns how many reads a complete certificate needs.

Symbol	Meaning
n_g, μ_g	Given cohort size and its positive-label rate.
N_h, K_h	Residual cell size and unknown positive count.
n_h, s_h	Formal records read from that cell and positives seen.
$[\ell_h, u_h]$	Current confidence interval for K_h.
$Z_j, [L_j, U_j]$	An inequality's score and its propagated range.
m_{hk}, c_{hk}	Extra reads in a prefix option; here $c_{hk} = m_{hk}$.
$\underline{C}, \overline{C}$	Checked lower cost and constructive upper cost.

For comparison, bounding the two rates separately gives $\mu_A \in [0.48, 0.60]$ and $\mu_B \in [0.35, 0.45]$, hence a looser difference $[0.03, 0.25]$. Combining the shared contribution first retains $[0.07, 0.21]$ and settles two questions in this example. For the three-item request, $\theta > 0.05$ is therefore true, $\theta < 0.06$ is false, and $|\theta| < 0.10$ remains unresolved. The request is incomplete until all three items are settled. These are schematic interval calculations, not measured acquisition savings.

For a general request, records with the same contribution to every requested comparison form one *cell*. Each cell needs one unknown positive count, not a separate count per comparison. Records whose net contribution is zero everywhere are removed from the effective support I_q. A charged uniform pilot is observed first; its known contribution becomes a constant, and the remaining records form the cells. Table I distinguishes the quantities used below.

C. How an Interval Settles a Decision

Call one strict inequality an *atom*. For nonempty cohorts and rational thresholds, including the pilot constant and multiplying by a positive common denominator gives every atom

an integer score

$$\mathcal{A}_j : \quad Z_j = \beta_j + \sum_h a_{jh} K_h > 0. \tag{2}$$

Scaling preserves the question and its boundary; cohort denominators stay fixed. The example's raw difference θ becomes such a score only *after* its threshold is subtracted and units are scaled.

To bound Z_j, add the smallest or largest possible contribution of every cell:

$$\begin{aligned} L_j &= \beta_j + \sum_h \min\{a_{jh}\ell_h, a_{jh}u_h\}, \\ U_j &= \beta_j + \sum_h \max\{a_{jh}\ell_h, a_{jh}u_h\}. \end{aligned} \tag{3}$$

Read this equation exactly as the endpoint calculation above. Since scores are integers, $L_j \geq 1$ proves the atom true; $U_j \leq 0$ proves it false. Otherwise more evidence is needed. Equality is false for a strict comparison.

A *witness* is a set of atoms sufficient for a positive decision. For practical equivalence, $|\theta| < \tau$ requires both $\theta + \tau > 0$ and $\tau - \theta > 0$. Its positive witness contains these two atoms; either false atom proves nonequivalence. For strict top-k, a candidate set S with $|S| = k$ is a witness if every member has a higher rate than every group outside S. A successful witness returns that set; no internal ordering is required. Ties can leave strict identification unresolved even after a census.

In general, let J_w be the atoms in witness w, and $\mathcal{W}_d$ the witnesses for decision d. The logical contract is

$$D_d = \bigvee_{w \in \mathcal{W}_d} \bigwedge_{j \in J_w} \mathcal{A}_j. \tag{4}$$

Here $\wedge$ means "all" and $\vee$ means "at least one." A true decision needs one all-true witness. For example, "either A or D beats both B and C has two witnesses: $(A > B, A > C)$ and $(D > B, D > C)$. A false decision needs a false atom in *every* candidate witness; an unresolved atom blocks neither route. This rule avoids explicitly listing every possible negative proof. Rankings allow only the positive, set-identifying side. A complete request needs a proof for every item.

We call this collection of supported atom assertions a *decision certificate*. It uses atom-by-atom interval checks. It does not perform every possible joint deduction: an interval spanning zero individually refutes neither $\theta > 0$ nor $\theta < 0$, although their conjunction is impossible. All cost bounds below refer to this stated certificate rule.

D. Where the Count Intervals Come From

The allocator chooses *which cell* to advance, not which unread record it will see. Each cell supplies the next records of an independently uniform random order without replacement. After n records with s positives, candidate totals lie between s and $s + N_h - n$.

The count ranges must remain trustworthy even when the next cell and stopping time depend on earlier observations. Conditional on the fixed labels, pilot and frozen cells, our construction guarantees that the probability of *any* cell range excluding its true count at *any* sampled prefix is at most α. On the coverage event every supported answer is correct for the chosen label operator. This concerns sampling uncertainty; it is not a classifier-accuracy guarantee.

The established finite-population confidence-sequence construction [11] rejects candidate totals that the observed sequence makes sufficiently implausible. Formally, its accepted candidates satisfy

$$\frac{q_h(y_{1:n})}{p_{h,K}(y_{1:n})} < \frac{1}{\alpha_h}, \qquad p_{h,K}(y_{1:n}) = \frac{(K)_s (N_h - K)_{n-s}}{(N_h)_n}. \tag{5}$$

Here $(x)_r = x(x-1)\cdots(x-r+1)$. The denominator is the probability of the *ordered* observed labels if the cell contains K positives. For example, from $N_h = 10, K = 3$, the sequence $(1, 0)$ has probability $(3/10)(7/9)$.

The numerator q_h is a working predictive probability, not extra label evidence. We use a half-and-half prior mixture of Beta$(1, 1)$ and a pilot-fitted Beta prior with 20 added pseudo-observations on top of Beta$(1, 1)$, giving total concentration 22. Its prior is frozen before formal sampling; conditional mixture weights update with observations. All policies share the same cells, priors and risk allocation, with $\sum_h \alpha_h \leq \alpha$. The exact mixture and likelihood calculation are given in Technical Companion TC1.

Accepted counts form an interval. After every record, the executor intersects it with the preceding interval, so evidence can only narrow. An empty intersection is a persistent error: later observations or a census cannot erase it. At exhaustion, the exact count is intersected with the history.

Conditional on the fixed labels, pilot and frozen cells, the construction gives

$$\Pr\{\forall h, n : \ K_h \in [\ell_h(n), u_h(n)]\} \geq 1 - \alpha. \tag{6}$$

On this event all returned decisions are correct. Reusing covered counts across many atoms spends no additional risk. The guarantee is per request and concerns the chosen labels, not their semantic quality. It holds for adaptive cell allocation because each next within-cell draw has the correct predictable null law. The companion gives the supermartingale argument; numerical implementation and observed exclusions are audited separately.

III. How Many More Reads Would Complete the Request?

A. Turning Interval Progress into Read Options

Suppose a particular positive witness still needs three units of improvement in one atom and two in another. Table II illustrates why shared acquisition matters: one read in cell 3 helps both atoms, whereas separate reads in cells 1 and 2 are needed for that witness. The full planner also considers alternative witnesses and false answers.

More generally, fix a current nonfailed state and a nested continuation of every cell's intervals. A continuation may be an actual path used offline or a hypothetical path used

TABLE II
SCHEMATIC ONE-READ OPTIONS FOR ONE POSITIVE WITNESS. REQUIRED GAINS ARE $(3, 2)$. CELL 3 ALONE SUPPLIES BOTH; CELLS 1 AND 2 TOGETHER COST TWO READS. THESE ARE ILLUSTRATIVE GAINS, NOT EXPERIMENTAL MEASUREMENTS.

Option	Cost	Atom 1 gain	Atom 2 gain
Read cell 1	1	3	0
Read cell 2	1	0	2
Read cell 3	1	3	2

for planning. Let $\underline{v}_{jh}(m)$ and $\overline{v}_{jh}(m)$ be the minimum and maximum contributions of cell h to atom j after m extra reads, using the min/max rule in (3). At $m = 0$ they are the current contributions. Define

$$g^+_{jh}(m) = \underline{v}_{jh}(m) - \underline{v}_{jh}(0),$$
$$g^-_{jh}(m) = \overline{v}_{jh}(0) - \overline{v}_{jh}(m). \tag{7}$$

These gains respectively raise an atom's lower endpoint and lower its upper endpoint. Nested intervals make both gains nonnegative and nondecreasing before a persistent failure. Their effects add across cells: $L_j(m) = L_j(0) + \sum_h g^+_{jh}(m_h)$ and $U_j(m) = U_j(0) - \sum_h g^-_{jh}(m_h)$. Thus the remaining gains needed for true/false proofs are $d^+_j = \max\{1 - L_j(0), 0\}$ and $d^-_j = \max\{U_j(0), 0\}$.

B. A Completion Program, One Rule at a Time

For cell h, prefix option k reads m_{hk} more records at cost $c_{hk} = m_{hk}$. Include the zero-read option. Binary x_{hk} selects an option. We minimize additional cost while choosing exactly one prefix in each cell:

$$\min \sum_{h,k} c_{hk} x_{hk}, \qquad \sum_k x_{hk} = 1 \quad (\forall h). \tag{8}$$

Evidence rule. Binary T_j or F_j selects an atom's true or false proof; both may be zero if the atom is unused. A selected side must receive enough gain:

$$\sum_{h,k} g^+_{jh}(m_{hk}) x_{hk} \geq d^+_j T_j,$$
$$\sum_{h,k} g^-_{jh}(m_{hk}) x_{hk} \geq d^-_j F_j, \qquad T_j + F_j \leq 1. \tag{9}$$

For example, the first inequality says that claiming a true atom activates its required gain; leaving it unused imposes no requirement.

Decision rule. Binary p_d, r_d choose a true/false decision proof, and z_w selects a positive witness. The logic follows Section II:

$$\begin{aligned} z_w &\leq T_j && (j \in J_w), \\ \sum_{w \in \mathcal{W}_d} z_w &\geq p_d, \\ \sum_{j \in J_w} F_j &\geq r_d && (w \in \mathcal{W}_d), \\ p_d + r_d &= 1. \end{aligned} \tag{10}$$

The first two lines select an all-true witness for a true answer. The third hits every witness with a false atom for a false answer. The last requires one justified value for each decision. For rankings, fix $r_d = 0$ so a selected set must be identified.

Proposition 1 (exactness). On fixed nonfailed nested paths, the completion program is feasible exactly when its selected prefixes complete every decision under the stated atomwise certificate rule.

Proof. A completed request supplies the asserted atom sides and positive witnesses or negative hits required by the program. The additive gains verify their evidence constraints. Conversely, each selected side has sufficient gain, and the decision constraints assemble those supported sides into a proof for every item. □

The representation is compact in its listed inputs: for H cells, M prefix options, J atoms, W witnesses, D decisions and E atom–witness incidences, it has $O(M + J + W + D)$ variables and $O(H + J + W + D + E)$ constraints. For disjoint two-atom witnesses, a negative-DNF expansion has 2^W terms while the negative-hit constraints have W clauses. Primary nonranking decisions have one witness and rankings forbid negative completion; their experiments test shared completion rather than this exponential representation advantage. General uncapped optimization is NP-hard by a Set Cover reduction given in the companion.

C. Reading a Cost Bound Correctly

The offline evaluator substitutes the *realized* interval paths, keeping every prefix before the first persistent failure. It also knows the fixed reference labels. To study correct proofs, it forbids false assertions on source-true atoms and true assertions on source-false atoms. This defines C^*_{src}, the least additional reads for a source-valid certificate. "Source-valid" means the selected assertions match that fixed label reference; it does not require every historical count interval to cover.

The oracle starts after the pilot. All reported bounds add its charge: $\underline{C} = |P| + \underline{C}_{\mathrm{add}}$ and $\overline{C} = |P| + \overline{C}_{\mathrm{add}}$. Here $|P|$ is the pilot size. A constructive upper bound is a checked allocation supplying a source-valid proof; a lower bound rules out all cheaper source-valid allocations under the same fixed paths, kernel, risks and atomwise rules. Compare these *total* costs with the total budget B:

- $\overline{C} \leq B$: a within-budget proof exists on those paths.
- $\underline{C} > B$: no source-valid atomwise completion exists within budget on those fixed paths.
- Otherwise: the bounds do not settle feasibility. For example, $[80, 110]$ cannot settle a budget of 100 reads.

For the same budget, a checked allocation costing 95 proves feasibility, whereas a lower bound of 105 rules it out. Every source-proof-valid completing policy reaches a feasible prefix allocation and therefore costs at least $|P| + C^*_{\mathrm{src}}$ in total. Batching and exploration can add cost. The evaluator can choose witnesses with hindsight; its feasible allocation alone is not a causal online policy. The companion also defines the unrestricted C^* for proofs without source-truth filtering and explains the distinction off the coverage event.

How the bounds are checked. Actual grid prefixes give feasible upper bounds. An optimistic grid supplies a later endpoint's gain at an earlier endpoint's cost, giving a lower relaxation. The optimistic integer optimum lower-bounds the full-prefix optimum. Moving each selected optimistic option to its later actual grid endpoint gives a feasible allocation, adding at most the sum of those grid widths. LP-based gaps additionally include integrality and solver gaps.

We do not accept a floating solver's bound as proof. Here x collects prefix and auxiliary variables, with zero acquisition cost on auxiliaries; encode any remaining equality in both inequality directions and include the source-truth fixes. Write these non-onehot constraints as $Gx \geq b$, convert candidate multipliers to nonnegative rational η, and form adjusted costs $\bar{c} = c - G^\top \eta$. Exact weak duality gives

$$\underline{C}_{\text{add}}(\eta) = \eta^\top b + \sum_h \min_{k \in h} \bar{c}_k + \sum_{v \in \text{aux}} \min\{0, \bar{c}_v\}. \quad (11)$$

Read the last two terms as the cheapest adjusted option in each cell and the cheapest $0/1$ value of each helper variable. Any such rational multipliers give a valid additional-cost lower bound; add the pilot for total cost. Upper allocations are checked separately with integer arithmetic. This is a certificate about *optimization cost*, distinct from an online decision certificate. Wide or unavailable bounds remain visible.

D. Separating Signed Overlap from Other Sharing

The example in Section II combined overlap before propagating intervals. To isolate this effect experimentally, we keep the paid reads, count intervals and support filtering fixed, but propagate each nonzero intersection term through separate left/right endpoints. For one unscaled contrast, this widens $[L, U]$ to $[L - P, U + P]$, where

$$P = \sum_{h \subseteq G_A \cap G_B \colon a_h \neq 0} \min\{1/n_A, 1/n_B\}(u_h - \ell_h). \quad (12)$$

In the running example, the intersection width is $12 - 8 = 4$, so $P = 0.01 \times 4 = 0.04$: $[0.07, 0.21]$ becomes $[0.03, 0.25]$. Both virtual appearances retain the *same* prefix and are charged once. Zero-net terms remain filtered and the pilot stays exact. A checked $\overline{C}_{\text{signed}} < \underline{C}_{\text{relaxed}}$ then proves that separate propagation needs strictly more evidence. It does not compare shared acquisition against two independently sampled systems.

E. Online Allocation and Forecast Planning

An online allocator sees only observed labels and frozen metadata. Forecast asks the completion program to plan against *predicted* future intervals, not the actual future used by the oracle. It centers each prediction at the pilot-mixture posterior-mean total, rounds half-up, clips to the current interval and predicts finite-population width shrinkage. Predictions are nested and end at the predicted singleton. They are scheduling estimates, never confidence intervals used for output.

A coordinate pass starts with the longest forecast prefixes and removes reads while keeping the forecast request complete. If even the longest prefixes fail, there is no forecast-feasible plan. An exactly checked, cheaper MIP solution may replace the coordinate incumbent. The retained plan advances the cell with the largest ratio of unfulfilled planned reads to unread population, then unfulfilled reads, then smallest cell ID. A plan-selected batch cannot exceed its remaining planned reads; common initialization/exploration takes priority. Fallback retains a usable plan or uses DifficultyGreedy.

The grid is $\{0, 16, 32, 64, 128, N_h - n_h\}$ after clipping/deduplication. Planning uses one thread, a 50 ms solver limit and 150 ms soft limit; it replans after eight rounds or when the plan is exhausted. The period was selected from $1, 4, 8$ on four earlier airline development queries. Exact shrinkage, rounding, ties and comparator scores are retained in Technical Companion TC2–TC3. Soft deadlines may change rerun actions.

Algorithm 1 shows how every policy uses the same evidence and stopping rule. Completion is checked after a successful batch, while intervals are updated after every record. All submitted attempts are charged even if an intermediate prefix would have sufficed. Failed/partial batches or a persistent interval failure suppress claims and retain charges; rendering failures also retain charges.

Algorithm 1 One shared executor; only the allocator changes

```
Compile questions; charge and observe a uniform pilot.
Freeze residual cells, priors, risks and uniform cell orders.

Set round t = 0 and count intervals [0, N_h].
loop
    if a persistent interval failure has occurred then
        Return error and submitted cost.
    end if
    Propagate count intervals; check each requested decision.
    if all decisions are justified then
        Return their values/sets, certificates and cost.
    end if
    if the budget is spent or no unread record remains then
        Return the unresolved items and cost.
    end if
    Increment t.
    if t is divisible by four then
        Explore a uniformly chosen nonexhausted cell.
    else if an unvisited cell remains then
        Choose largest cell, ties by largest cell ID.
    else
        Ask the selected allocator for a cell and batch.
    end if
    Use 8 reads on a first visit, otherwise at most 16 (Forecast may use fewer); clip to unread records and budget.
    Charge and acquire the next batch of that cell's order.
    Update its count interval after each returned label.
end loop
```

IV. Experimental Design

The studies answer different questions and use different cost units (Table III). A saved-label replay counts which labels a policy would request; it is not a new model forward. The offline oracle analyzes selected replay paths with hindsight. The MAST study reads a whole vector at once. Only the live study measures new physical model calls and time.

A. Success, Cost and Uncertainty

Our primary success criterion is deliberately whole-request: every requested value and every atom used in its certificate must match the fixed full reference. A correct answer with a wrong selected proof does not qualify. An interval excluding a true cell count is a separate event; it need not make the final answer wrong. Unresolved requests, wrong claims, persistent interval failures and execution errors remain in all planned denominators.

Primary cost is submitted labels divided by distinct effective-support size $|I_q|$, averaged over *all* planned executions, including early failures. It is not a success-conditioned time or budget fraction. The common cap is $B = \lceil 0.5|I_q| \rceil$, so 0.25 labels/support uses about half that cap. A sensitivity charges errors and invalid proofs to the cap. Savings among jointly completed pairs are secondary and state their selection denominator.

Ten whole-workload seeds supply independent blocks; methods share their pilot, cell-order, exploration and planning streams. Shared rows, reversed comparisons and common templates make executions within a block dependent. Paired 95% percentile intervals resample these ten *blocks* $20{,}000$ times, conditional on the fixed source/workload. Balanced summaries weight four predicates and three families equally: pairwise, equivalence and three-decision summaries (pooling equal-sized dominance/mixed groups). The eight MAST paths are reported descriptively instead of using these intervals.

An analyzer independent of the executor/compiler reconstructs reference means and decision families, checks selected proof atoms and costs, and scans available count paths. It distinguishes a reference exclusion, a wrong claim, a persistent interval failure and a framework error. Preselected evaluations and outcome-selected adverse diagnostics keep separate denominators.

B. CivilComments: Natural Overlap and Two Label References

A salted metadata hash selects $30{,}000$ eligible rows from the identity-enriched CivilComments test split distributed with WILDS [12]. Eligibility requires at least one of eight existing content-mention scores ≥ 0.5. Selection uses no target label or outcome, and duplicate texts retain distinct source IDs. These cohorts describe what a comment mentions, not the commenter's identity or a representative demographic population.

The cohort sizes are 2,011–9,409; all 28 intersections are nonempty (98–3,291 rows). Multiple memberships occur in $8{,}767$ rows (29.22%), as shown in Fig. 2. Four existing annotation scores—toxicity, insult, identity attack and obscene language—become binary crowd references at decimal score ≥ 0.5, including ties.

For each predicate, requests include 56 directed comparisons, 28 unordered equivalence judgments (tolerance 0.05), 16 dominance summaries, 16 mixed summaries and 16 four-cohort rankings (7 top-1, 9 top-2). A dominance summary asks three comparisons against one anchor; a mixed summary asks one comparison and two equivalences over six distinct cohorts. Across four predicates, there are 464 primary nonranking and 64 secondary ranking contracts. With ten seeds, the primary denominator is $4{,}640$ executions per method/reference; the summary denominator is $1{,}280$. No truth or certifiability filter is applied.

A separate $30{,}000 \times 4$ matrix supplies Qwen3-8B labels [13] on the *same texts*. The pinned operator fixes prompts, weights/tokenizer, BF16, SDPA, a nonthinking template, one record per forward and next-token $0/1$ scoring (ties positive). Generation and KV/result reuse are disabled; all contexts fit $4{,}096$ tokens without truncation. Canaries check operational stability separately. Primary model-reference acquisition replays this saved matrix. Matrix construction, live-query timing and crowd/model semantic disagreement are distinct measurements.

C. GoEmotions: A Second Source

The three checksum-verified GoEmotions release files [14] contain $58{,}011$ distinct IDs. Of these, $57{,}877$ have at least two clear, distinct raters; a metadata hash selects $30{,}000$ without using emotion scores or outcomes. Anger, joy, sadness and fear are positive when at least half the fixed clear raters select them, including ties.

The eight metadata-only cohorts are the four most frequent subreddits *among the selected rows*, length/time at least their medians on those same rows, and question/exclamation marks. In that order, sizes are $139, 125, 122, 122, 15{,}070, 15{,}000, 3{,}044, 4{,}145$; exact names and cutoffs are in the public metadata. The same 132 cohort-index templates transfer to each emotion, with ten fresh seeds. This tests transfer across sources using the same request design and crowd labels only.

D. Common Execution and Allocation Policies

Request risk is $\alpha = 0.05$, allocated equally across residual cells with downward rounding so their risks sum to at most α. The charged pilot has size $\min\{256, \lfloor |I_q|/10 \rfloor, \lfloor B/4 \rfloor\}$. Label-free TF–IDF/SVD-128 features and pilot-only logistic regression supply a frozen proxy; a smoothed pilot probability handles empty/single-class or failed fits. If $\bar{p}_h$ is a cell's mean proxy probability, its Beta shapes are $1 + \lfloor 20\bar{p}_h \rfloor$ and $21 - \lfloor 20\bar{p}_h \rfloor$. There is one cell per coefficient signature. All policies use the same eight-read initialization, sixteen-read batches, every-fourth-round exploration, intervals and checker. Policies choose the next cell; Forecast may also clip its batch to the retained plan.

TABLE III

HOW TO READ THE EVIDENCE. AN EXECUTION IS ONE REQUEST, SEED, REFERENCE AND POLICY COMBINATION; A REQUEST MAY CONTAIN SEVERAL DECISIONS. THE MATHEMATICAL CONSTRUCTION IN SECTION VII IS SEPARATE FROM ALL MEASURED STUDIES.

Study	Question	Declared scope	Charged or measured unit
Allocation replay	Does allocation improve complete-request output?	73,760 planned executions	One submitted record–predicate label attempt.
Offline bounds	Does a proof exist within budget on the same paths?	192 preselected cases	Total reads, including pilot; checked cost brackets.
Earlier-corpus replay	Do patterns recur on earlier saved labels?	9,600 executions	Record–predicate label attempts.
MAST programs	How do configured inference architectures behave?	7 programs, 8 paths each	One record revealing all 14 released flags.
Live inference	How do acquisitions map to actual model costs?	23 query–policy executions	Physical forwards, formatted tokens and wall time.

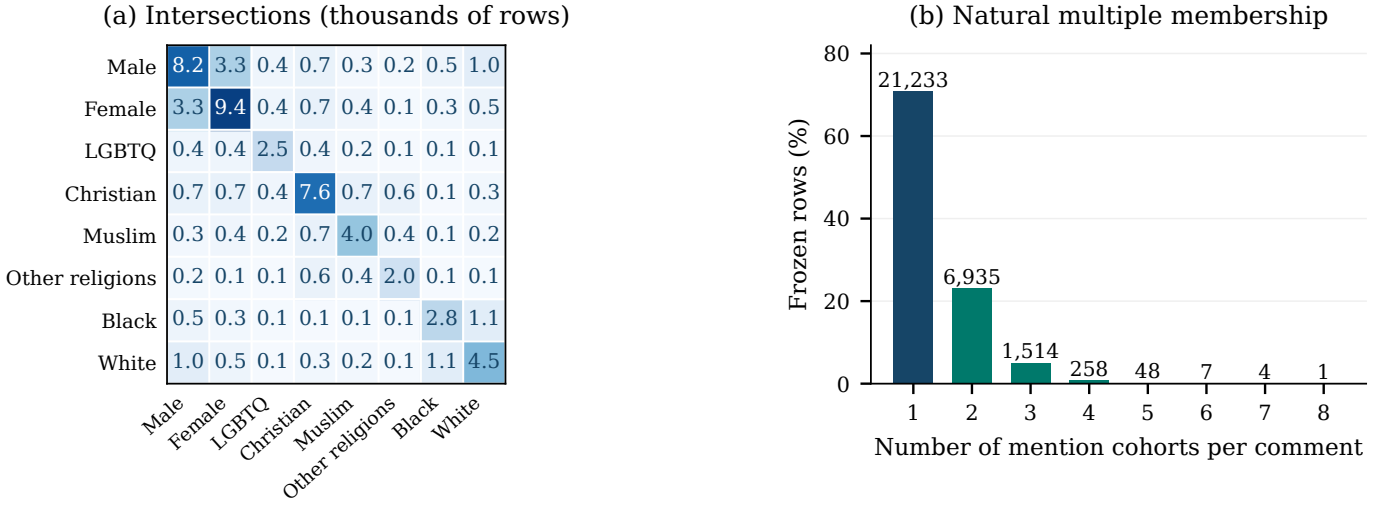


Fig. 2. Metadata of the 30,000 selected CivilComments rows. In (a), diagonal values are cohort sizes and off-diagonal values are intersections, rounded to thousands; exact integers accompany the artifacts. Panel (b) counts each row once. Memberships use existing content-mention scores, and neither plot uses target labels.

Matched balances the number of formal records read: choose the cell with smallest n_h, then smallest ID. *Width-Greedy* chooses the greatest predicted reduction in normalized live-contrast width per read. *VarianceGreedy* instead targets estimated variance reduction with the same pruning and normalization, adapting Neyman allocation [15]. Settled decisions and refuted positive witnesses are pruned. Technical Companion TC3 supplies exact normalization, shrinkage and tie formulas for these controls.

DifficultyGreedy targets progress toward a complete certificate. To make its score interpretable, let an atom's original contrast interval have width W, midpoint m and threshold b_0. For the desired proof side, let e be the signed distance of the midpoint from that threshold. Its difficulty is

$$r = \frac{W}{2\max\{e, \varepsilon\}}, \qquad \varepsilon = 10^{-4}. \tag{13}$$

A wide interval or a midpoint near the boundary is difficult; the opposite proof side uses $-e$. An all-true witness combines unresolved atom difficulties with log-mean-exp, a smooth maximum. Choose the easiest positive witness. A false proof first chooses the easiest false atom per surviving witness, then combines those difficulties across witnesses. A binary decision uses the easier proof side; ranking uses only the positive side. Sum across unresolved decisions to obtain $\mathcal{D}$. Resolved decisions contribute zero; blocked rankings remain incomplete.

For each candidate cell, predict width shrinkage while keeping midpoints and witness eligibility fixed. With batch size b_h clipped to 16, unread records and remaining budget, choose

$$h^* = \arg\max_h \frac{\mathcal{D}(S) - \mathcal{D}(\widetilde{S}_h)}{b_h}, \tag{14}$$

where S is the current state and $\widetilde{S}_h$ the predicted state after that batch. This chooses the greatest reduction in remaining proof difficulty *per paid read*. Gain ties use smallest cell ID within the frozen tolerance; nonpositive gain falls back to largest N_h, then smallest ID. The common checker, not this score, decides completion. TC3 gives the full equations, including log-mean-exp, pruning and exceptional states; this is our internal heuristic, not Adaptive Dual Greedy [5].

For one scalar comparison, the easier-side difficulty is $W/[2\max\{|m - b_0|, \varepsilon\}]$. Under midpoint-fixed shrinkage, the

denominator is common to all cells. DifficultyGreedy therefore orders candidate batches exactly as width reduction in real arithmetic, including clipped sizes. Single-contrast equivalence has the same width ordering for equal-price candidates; floating ties may differ. This explains why the stronger width control is necessary.

MaxDifficulty replaces the sum of cross-decision marginal gains by their maximum on the 128 three-decision summaries. Shared labels and inference stay fixed, so this isolates aggregation, not independent acquisition. *Forecast* uses Section III-E. For CivilComments rankings, *Boundary* chooses a tentative top-k by relative plug-in means, then targets the inside group's smallest relative lower bound against the outside group's largest relative upper bound. Its exact origin/tie/fallback rules are in TC3; it is a configured adaptation.

The total replay count is 31,680 original CivilComments executions, 1,280 ranking Boundary executions, 23,680 stronger-control executions and 17,120 GoEmotions executions: 73,760 in all. The original population, queries, operator and three policies were internally hash-frozen before target conversion/inference. Stronger controls are post-hoc to those outcomes but code-frozen before their runs. GoEmotions inputs/policies were frozen before target conversion, after CivilComments results were known. These are internal freezes, not external preregistration.

E. Offline, Vector-Label and Live Studies

Offline cases. Support strata and salted hashes select 48 nonranking IDs across four predicates, four subtypes and three strata, without outcome selection. Two seeds and two CivilComments references give 192 cases, paired to the original three policies. Full paths, source-valid atom restrictions and risks match. Missing or wide bounds remain. Relative gap is $(\overline{C}-\underline{C})/\max\{1,\overline{C}\}$ on total costs. The endpoint intervention uses the same cases/paths with linked single-charge appearances; exact arithmetic checks its bounds. Policy headroom compares summed cap-charged policy costs to summed oracle costs capped at each budget; the full deterministic bracket formula and solver limits are in TC3. No online policy is rerun for this intervention.

MAST programs. One fixed request uses released multi-agent traces from MAST-Data [16]: 100 each from MetaGPT/GPT4o (A), MetaGPT/Claude (B) and ChatDev/GPT4o (C). One acquisition reveals all 14 released binary judge flags. Completion requires all 28 values $\mu_{A,j} > \mu_{B,j}$ and $\mu_{A,j} > \mu_{C,j}$; ties are false. These judge labels are the reference, not human gold.

Seven configured programs each have eight planned trajectories; budgets $75, 150, 225, 300$ are readouts of those same trajectories. The local RR/Width/DWidth programs share 42 marginal count sequences, five charged pilots per cohort and risk $1/840$ per stream. NativeExact uses Evergreen's seeded exact aggregates [4]; NativeProgressive uses synchronized full-population intervals and persistent intersections. Joint RR/LogGreedy use predictable without-replacement Kelly bets and verified composite-null witnesses [9], with risk $1/560$ per comparison shared by its two signs. Their cohort prefixes may differ. Exact kernels, predictions and native settings are in TC3.

An MAST path earns credit only after verified normal closure, known cost within budget, all 28 correct supported outputs, and no earlier source-count exclusion or erased error. Interrupted/unverifiable paths are uncredited, with unknown cost tails left unknown. All four censors, including two Native startup censors, remain; there are no retries. This historical-evidence condition is additional to the primary endpoint. Vector reads are not model forwards, tokens or monetary costs.

Replications and live queries. Earlier sources have 40,000 CFPB narratives [17], [18] and 18,569 OpinRank reviews [19], four saved predicates each. Their combined 480 contracts split $80/80/320$ into train/development/test. The 320 test contracts, three policies and ten seeds yield 9,600 revised replays with 50% caps. These are post-hoc, previously examined labels with an unpinned upstream model and no human validation of the eight predicates.

The live schedule selects five CivilComments queries and one 14,640-row airline ranking [20] by metadata hashes before method-outcome inspection; primary crowd results were already known. Six queries times three policies plus Boundary on two rankings give 20 warm executions. Three fresh-process executions use the same CivilComments toxicity pairwise query, one per primary policy. Forwards, tokens, planning and lifecycle time are recorded separately. The public repository provides fixed inputs, analysis/method sources, bound checks, runtime receipts and the complete Technical Companion; its notices state external data/model requirements.

V. Results

A. Allocation Helps Most on Multi-Decision Requests

The strongest matched control leaves a modest allocation gain. On CivilComments, DifficultyGreedy completes 36 more crowd-reference and 54 more model-reference requests than WidthGreedy, out of 4,640 units each (Table IV). The gains are 0.78 percentage points (95% block interval $[0.69, 0.86]$) and 1.16 points ($[0.91, 1.44]$). Relative reductions in mean all-unit labels/support are 0.54% and 0.60%. Against VarianceGreedy, completion gains are 3.25 and 5.41 points, with 2.89% and 3.94% fewer labels. Against Matched, the reductions are 5.51% and 6.04%. These comparisons hold shared evidence, support filtering and inference fixed.

The gains come from summaries with several decisions. DifficultyGreedy and WidthGreedy agree on calls, status, completion, answer/proof correctness and errors in all 6,720 paired CivilComments comparison/equivalence units. This observable equality need not imply identical read order. It agrees with the scalar reduction in Section IV. Figure 3 shows complete-request stopping costs and retains every unsuccessful unit in the denominator.

On GoEmotions, DifficultyGreedy and WidthGreedy nearly tie: 2,896 versus 2,894 completions of 4,640, a 0.043-point gain with interval $[-0.065, 0.129]$ and 0.20% fewer labels. Against VarianceGreedy, the gain is 2.67 points with 2.98%

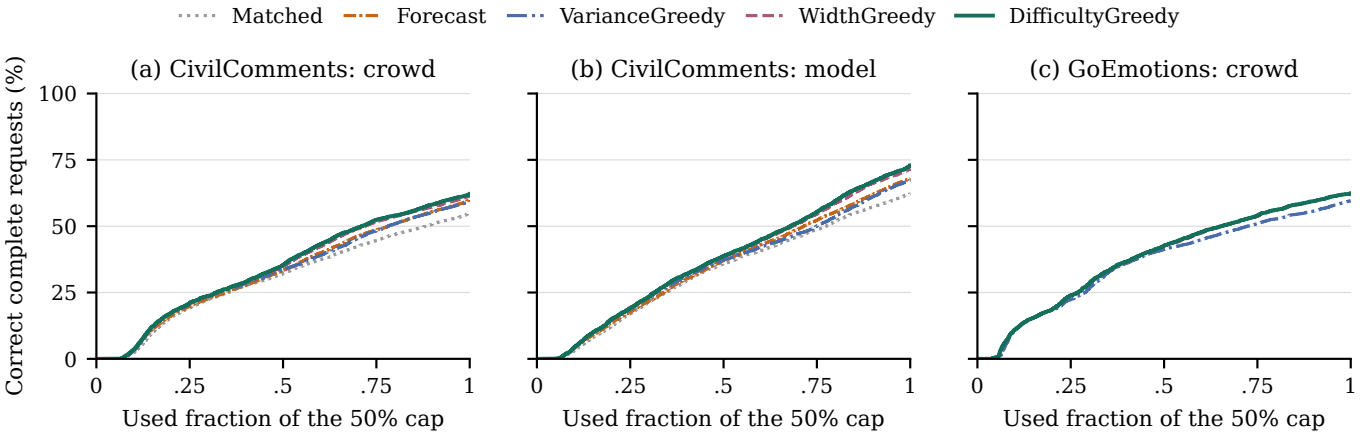


Fig. 3. DifficultyGreedy has a small advantage over WidthGreedy on CivilComments and nearly ties it on GoEmotions. Curves show correct reference-supported completion over all 4,640 planned primary nonranking executions per allocator/reference. Horizontal position is the used fraction of the 50% cap: 1 is approximately 0.5 labels/support. Curves describe fixed runs, not restarted executions at each budget. Errors, unresolved requests and wrong proofs remain uncompleted. CivilComments references share texts; GoEmotions is a second source.

TABLE IV
ALL PRIMARY NONRANKING UNITS: 4,640 PER ROW. CIVIL C/M: CIVILCOMMENTS CROWD/MODEL; GO C: GOEMOTIONS CROWD. CORRECT REQUIRES EVERY VALUE AND SELECTED PROOF TO MATCH ITS REFERENCE. CALLS/$|I_q|$ AVERAGES ALL UNITS. E/W COUNTS EXECUTION ERRORS/SOURCE-WRONG PROOFS, RETAINED AS UNSUCCESSFUL.

Reference	Allocator	Correct	Calls/$\lvert I_q \rvert$	E/W
Civil C	DifficultyGreedy	2,886	0.3265	8/0
Civil C	WidthGreedy	2,850	0.3283	9/0
Civil C	VarianceGreedy	2,735	0.3363	10/0
Civil C	Matched	2,548	0.3456	28/0
Civil C	Forecast	2,784	0.3365	10/0
Civil M	DifficultyGreedy	3,384	0.3125	24/1
Civil M	WidthGreedy	3,330	0.3144	24/1
Civil M	VarianceGreedy	3,133	0.3253	22/0
Civil M	Matched	2,892	0.3326	37/1
Civil M	Forecast	3,158	0.3242	20/1
Go C	DifficultyGreedy	2,896	0.3084	3/0
Go C	WidthGreedy	2,894	0.3090	3/0
Go C	VarianceGreedy	2,772	0.3178	5/0

fewer labels. All 3,360 paired comparison/equivalence units agree on measured outcomes. On summaries, sum versus maximum gain completes 837 versus 833 of 1,280: 0.31 points ($[-0.08, 0.70]$) and 0.47% fewer labels. Across all 17,120 GoEmotions executions the audit retains 14 persistent interval errors and no source-wrong proof. Secondary ranking counts are 317, 312 and 256 of 640 for DifficultyGreedy, WidthGreedy and VarianceGreedy. Thus the second source supports the evaluation procedure and the scalar equality, while its sum-aggregation advantage is uncertain.

Forecast improves on Matched overall but loses 2.20 crowd-reference and 4.87 model-reference completion points to DifficultyGreedy on CivilComments, while using more labels. Its crowd summaries complete 695 of 1,280, versus Matched's 432. Yet its comparison completions fall to 1,317 of 2,240 (Matched: 1,340), and equivalence completions to 772 of 1,120 (Matched: 776). More elaborate planning need not improve every family. Equal weighting of predicates and families preserves the main ordering.

B. What Gain Aggregation and Signed Overlap Change

Summing progress across decisions. On 1,280 CivilComments summaries per reference, DifficultyGreedy's summed gain completes 746 crowd and 907 model units. Replacing the sum by a maximum completes 728 and 894. The respective gains are 1.41 points ($[0.94, 1.95]$) and 1.02 points ($[0.39, 1.72]$), with 0.84% and 1.29% fewer normalized labels. All crowd gains are dominance summaries; model gains comprise eleven dominance and two mixed summaries. Both policies already share labels and inference. This comparison isolates aggregation across decisions.

Combining overlap before interval propagation. All 192 fixed-endpoint cases reconcile with physical prefixes, single-charge reads and rational cost checks. In 13 of 96 crowd cases and 17 of 96 model cases, the signed formulation's checked upper cost lies below the relaxed formulation's lower cost. Separate propagation therefore requires strictly more evidence under the stated checker. Within these subsets, median lower bounds on extra reads are 91 for crowd and 42 for model labels. The remaining cases do not prove a strict cost difference; no pair proves a switch across the 50% cap. No online policy is rerun for this intervention.

The relaxed study has 191 finite brackets and one case with no source-valid proof. Relative gaps have medians 7.62% for crowd and 7.16% for model labels, with maxima 54.40% and 61.37%. These uncertainties limit the number of strict comparisons the intervention can establish.

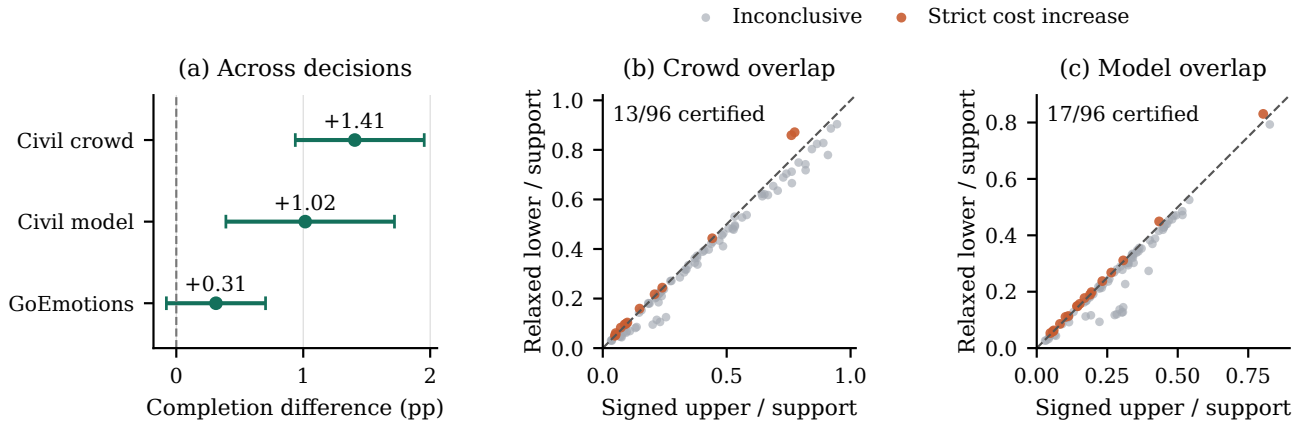


Fig. 4. Two mechanisms with evidence held fixed. (a) Sum versus maximum aggregation on 1,280 summaries/reference; bars use paired 95% seed-block intervals. (b,c) Signed upper costs versus relaxed lower costs, both including pilot and normalized by support. Points above equality prove an extra necessary cost; gray points are inconclusive. Panels (b,c) show finite cost pairs. The model case with no legal source-valid proof stays in the 96-case denominator and Table V, but has no finite scatter coordinates. No case proves a crossing of the 0.5 cap.

TABLE V
ALL 192 PRESELECTED OFFLINE CASES. THE FIRST FOUR ROWS PARTITION EACH REFERENCE'S 96 CASES. ONLINE COMPLETION COUNTS BELOW THE RULE ARE SEPARATE OBSERVATIONS, NOT AN ADDITIONAL PARTITION. ALL COSTS INCLUDE THE PILOT.

Diagnostic or policy	Crowd	Model
Within-budget proof constructed	67	90
Lower bound exceeds budget	24	3
No source-valid proof on any legal prefix	0	1
Bounds do not settle the budget	5	2
DifficultyGreedy completes	62	82
Forecast completes	61	75
Matched completes	52	66

C. Offline Bounds Separate Opportunities from Barriers

Table V answers a different question from online completion: does a source-valid proof exist within budget on these realized evidence paths? Constructive allocations answer yes in 157 of 192 cases. Verified lower bounds rule out a within-budget proof in 27; another model case has no source-valid proof at any legal prefix. Seven cases remain undetermined. All cases reconcile with the three methods' initialization hashes, IDs, labels and cell endpoints; none is replaced.

DifficultyGreedy leaves 13 cases uncompleted despite a constructed within-budget proof; Forecast leaves 21. These are hindsight opportunities under the fixed rules. They do not establish that an online policy could identify the right prefixes and witnesses without knowing future labels. Many crowd noncompletions instead face proved cost barriers. Figure 5 shows both kinds of evidence without collapsing uncertain bounds to a point estimate.

Of 191 finite signed brackets, 103 reach a 5% relative total-cost gap; the other 88 reach gaps up to 63.14%. A precision pass over all cases retains rational dual multipliers without changing case selection or upper proofs. Relative to DifficultyGreedy's cap-charged summed calls, bounded hindsight headroom is 7.03–12.59% on crowd labels and 13.55–23.97% on model labels. This brackets opportunity rather than guaranteeing attainable online savings.

D. Planning Diagnostics and Additional Workloads

Better forecast plans can still acquire poorly. At 78,937 observed crowd planning states, MIP finds a cheaper forecast-feasible plan than the coordinate incumbent. These are improvements to hypothetical plans, not verified acquisition savings or an isolated causal effect of MIP. Mean saved-label replay CPU time is 1,272 ms for Forecast, 608 ms for DifficultyGreedy and 531 ms for Matched; these are not live-model timings.

A post-hoc check compares the first fresh feasible forecast's total predicted cost P with a verified oracle lower bound L. In 9 of 94 crowd and 6 of 91 model cases whose full count paths cover their reference, $P < L$ proves optimism. All 192 cases have plans; the two crowd and five model source-excluding paths remain separate. Median oracle gaps in the covered subsets are 4.92% and 4.98%. A prediction above a loose lower bound is inconclusive. This diagnostic does not establish why the policy ordering occurs.

Earlier corpora. All 1,600 units per method/corpus remain. On CFPB, Matched, DifficultyGreedy and Forecast complete 653, 721 and 701, with mean labels/support 0.3920, 0.3805 and 0.3865. On OpinRank they complete 228, 326 and 298, with costs 0.4838, 0.4780 and 0.4800. Independent source/prefix/cost/proof audits reconcile all 9,600 executions, including an adverse path described below. These revised bidirectional, all-unit results are distinct from historical one-sided, certifiability-conditioned results.

Secondary rankings. Of 640 CivilComments crowd ranking units, DifficultyGreedy, WidthGreedy and VarianceGreedy complete 353, 324 and 312; on model labels they complete 454, 422 and 392. Forecast, Matched and Boundary complete

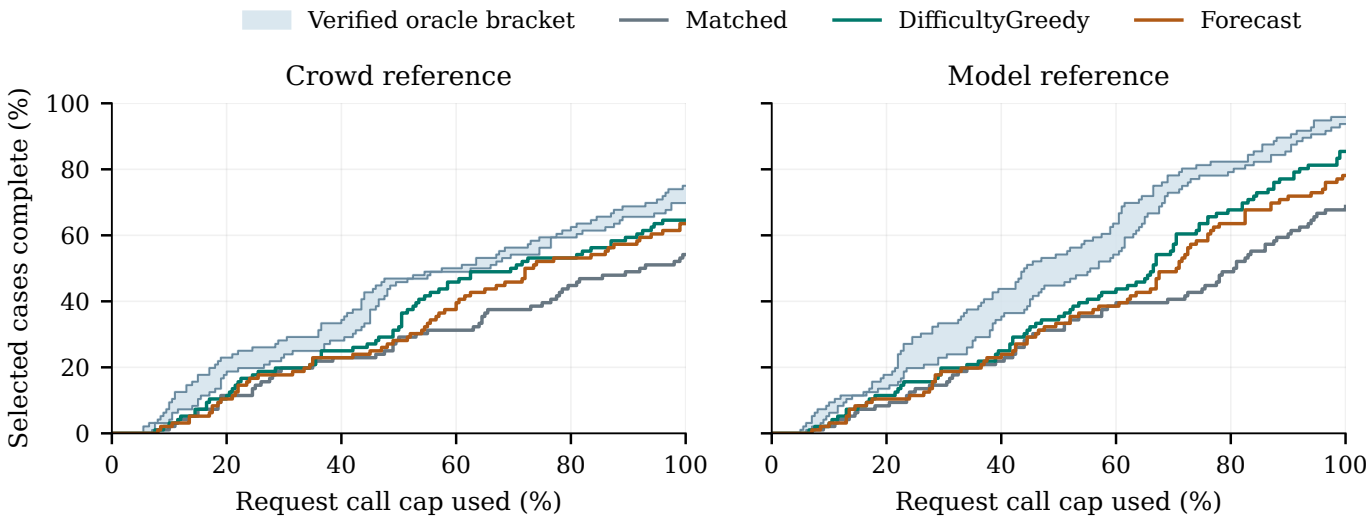


Fig. 5. Offline source-valid proof costs, 96 cases/reference. Horizontal position is percentage of the declared call cap. The band's lower edge counts constructed proofs within each budget; its upper edge additionally counts cases whose bounds still permit completion. All cases stay in the denominator. The band shows deterministic cost-bound uncertainty, not a statistical confidence interval. Policy curves are fixed-run stopping costs with unsuccessful cases retained. The offline evaluator uses future paths under the fixed kernel, risks and checker.

TABLE VI
PINNED MODEL LABELS RELATIVE TO EXISTING THRESHOLDED CROWD REFERENCES, ALL 30,000 ROWS PER PREDICATE. PERCENTAGES ARE DESCRIPTIVE; PRECISION/RECALL CONCERN THE POSITIVE CLASS.

Predicate	Agreement	Precision	Recall	Bal. acc.
Toxicity	83.03	48.16	36.79	64.48
Insult	74.86	19.65	70.81	73.01
Identity attack	74.39	17.25	69.46	72.11
Obscene	96.58	9.30	48.72	72.81

342, 298 and 295 crowd units and 431, 338 and 355 model units. These comparisons use the scoped Boundary adaptation; they do not establish superiority over canonical LUCB, IFocus or a full semantic engine.

E. Sampling Guarantees and Label Meaning Are Different

The original three-policy CivilComments grids retain 50 crowd and 100 model persistent interval errors. Three source-wrong model false-equivalence outputs arise from one shared query/seed path. The additional controls retain their errors and wrong proofs (Table IV); no favorable retry is taken. One shared CFPB path also has wrong values and proofs. Exact-rational checks permit the CivilComments and CFPB exclusions at nominal thresholds, without identifying a numerical defect. A count exclusion, a persistent interval failure, an unresolved output and a wrong proof are different events. These dependent paths do not calibrate empirical risk.

Model-relative validity also does not imply agreement with crowd labels. Across 30,000 records per predicate, agreement ranges from 74.39% to 96.58%, while positive precision ranges from 9.30% to 48.16% (Table VI). Prompts paraphrase the source concepts, not identical historical questionnaires. Canaries check operation and stability; high agreement on a rare label need not mean good positive classification. Artifacts include confusion counts, prevalence, recall, balanced accuracy and mention-cohort breakdowns.

Before sampling, the two full references already differ on 108 of 464 nonranking contracts and on the selected sets for 13 of 64 rankings. Of DifficultyGreedy's 3,838 model-proof-valid completions across all 528 contracts and ten seeds, 747 have a crowd-inconsistent answer or selected proof (counting their union). Of 5,250 model-reference runs with a recorded terminal interval collection, 2,027 have at least one crowd contrast outside its model-target interval. These are alternate-reference disagreements, not extra noncoverage of the fixed model target. The eight predicates in the earlier corpora remain without human-reference validation.

F. One Vector-Valued Request Across Architectures

This study charges one acquired record for all 14 released flags, unlike the scalar-label studies above. Table VII retains all seven configured programs and eight paths for the same 28-comparison request. None earns whole-request credit at budgets 75, 150 or 225.

MAST's DWidth is distinct from the primary DifficultyGreedy. Width and DWidth have identical counts and costs on all eight paths, so difficulty weighting adds no measured benefit here. Joint–LogGreedy's conditional cost range on seven paths cannot establish unconditional savings over Joint–RR's eight completed paths.

A post-hoc import-context correction reverified the unchanged trajectories. NativeProgressive earns $5/8$: two normally closed paths have correct final vectors but exclude a true count earlier. For population size 100, 99 reads and 19 observed positives, the true total 20 is lost when binary64 clipping/count readout changes the previous $[19, 20]$ interval to an upper count of 19.

TABLE VII
MAST FIXED REQUEST. COMPLETE: CREDITED PATHS AT BUDGET 300. COST RANGES CONDITION ON CREDITED PATHS. OTHER PATHS REMAIN IN THE EIGHT-PATH DENOMINATOR; INTERRUPTED/UNVERIFIABLE COSTS REMAIN UNKNOWN.

Program	Complete	Cost	Other paths
Local-CS / RR	7/8	298	1 censored
Local-CS / Width	8/8	278–288	None
Local-CS / DWidth	8/8	278–288	None
NativeExact adapter	7/8	300	1 censored
NativeProgressive / RR	5/8	300	2 prior exclusions; 1 censored
Joint-Kelly / RR	8/8	298	None
Joint-Kelly / LogGreedy	7/8	278–290	1 censored

TABLE VIII
LIVE COSTS: TOTAL FORWARDS AND MEAN SECONDS OVER SIX WARM QUERIES PER POLICY, INCLUDING UNRESOLVED ONES. FRESH-PROCESS QUERY/LIFECYCLE SECONDS USE ONE COMMON CIVILCOMMENTS QUERY; LIFECYCLE INCLUDES STARTUP AND EXIT.

Policy	Warm total forwards	Warm mean seconds	Fresh process query/lifecycle (s)
Matched	21,903	94.78	40.153/66.254
DifficultyGreedy	18,125	78.68	37.687/64.455
Forecast	21,039	91.90	40.300/68.708

A separate arithmetic audit finds such interface losses in 94 of $5{,}049$ enumerated proper-prefix states with population size 100; directed enclosures remove those enumerated losses. This is not an empirical error-rate estimate. No confidence interval or method trajectory is rerun, and the original records remain available.

G. What the Acquisitions Cost in Live Inference

The 23 live executions use one RTX 5090 (32 GB) with the pinned Qwen3-8B operator. They produce $69{,}799$ verified corpus forwards and $13{,}302{,}613$ formatted tokens, plus 192 setup-canary forwards and $17{,}888$ tokens. Independent checks verify proofs and $63{,}911$ interval updates.

Correct completion is $17/20$ warm and $3/3$ fresh-process. All three unresolved warm policies exhaust the same insult-equivalence query's $2{,}766$-read cap. Table VIII includes them in warm means; Boundary's two rankings are separate.

Across the five jointly correct warm query pairs, summed wall-time ratios are 1.2410 for Matched/DifficultyGreedy and 1.0356 for Matched/Forecast. This selected-pair comparison is secondary. Warm query time totals $1{,}703.034$ s; enclosing lifecycles total $1{,}731.961$ s, including setup, I/O and teardown. Single measurements occupy three chronological windows. Fresh processes may use warm filesystem caches; feature preparation is separate.

One exactly checked count exclusion recurs in three warm and three fresh-process dependent paths, while their final proofs remain source-valid. Overall, 20 of 23 executions complete correctly. Six full-support materialized lookups use no new forwards; without a controlled cold-build ledger, these timings do not establish when materialization amortizes its construction cost.

VI. RELATED WORK

Semantic queries. LOTUS, ZenDB, DocETL and Bond optimize model operators [1], [21]–[23]; ABAE, InQuest, Thalamus and SUPG study expensive-predicate approximation [2], [3], [24], [25]. Evergreen verifies aggregate/ordinal claims [4]; SemViews separates prior-label planning from fresh certification [26]. Our focus is complete comparative requests and their proof costs.

Acquisition and inference. Boolean evaluation [5] and decision diagrams [7] assume known distributions; query-guided resolution uses Boolean provenance [6]. Verification-error models guide re-verification [27]. PEAK pools stream capital [8]; stratified/target-aware inference handles composite objectives [9], [10]. Model confidence sets use joint regions and target-dependent intersections [28]; constrained selection projects risk rectangles onto optima [29]. Our frontier fixes one interval proof architecture.

IFocus, LUCB and best-arm/multiple-answer identification target unresolved orderings or top-set boundaries under their respective assumptions [30]–[33].

VII. DISCUSSION AND SCOPE

When the inference rules matter. A separate construction uses three disjoint 100-record cohorts A, B, X, 14 flags and a 15-read pilot. It asks 28 comparisons $\mu_{i,m} > \mu_{X,m}$ ($i = A, B$; $m = 1, \ldots, 14$), where $\mu_{i,m}$ is flag m's rate in cohort i. With risk $\alpha = 0.05$, all-zero hidden labels and the specified 42-stream local family, at least 205 vector reads are necessary. An OR-composite proof of the same request needs at most 197. The 171-read variant changes the request to $\mu_{i,m} - \mu_{X,m} > 0.05$ and uses $\mathrm{Beta}(1, 1)$; risk remains 0.05. TC2 derives these constructed costs; they are not MAST measurements.

What can transfer between architectures. TC2 diagnoses one displayed without-replacement transfer [34]; the published with-replacement result and other joint approaches remain intact. Frontier costs condition on paths, partition, kernel, risks and atomwise rules.

How to use the diagnosis. Hindsight feasibility suggests improving allocation; a proved barrier motivates revisiting inference or budget. Neither supplies a causal policy. Scope is eight cohorts, shared templates, one MAST request and 23 live executions. Semantic fidelity, drift and session-wide guarantees require additional assumptions. Verification-error models [27] and batching/materialization [23], [26], [35] address complementary deployment costs.

VIII. CONCLUSION

CohortCert connects complete comparative requests to shared evidence and checkable proof costs. Its compiler, executor and offline bounds distinguish allocation opportunities from barriers under stated rules. Experiments quantify signed-overlap effects and allocation gains while separating hypothetical plans, actual execution and label meaning.

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